

REAR WING & SUSPENSION DESIGN: FINAL REPORT



Author:	Sam O'Neill
CID:	00729321
Project Supervisor:	Dr. Joaquim Peiro
Academic Supervisor:	Dr. David Moxey
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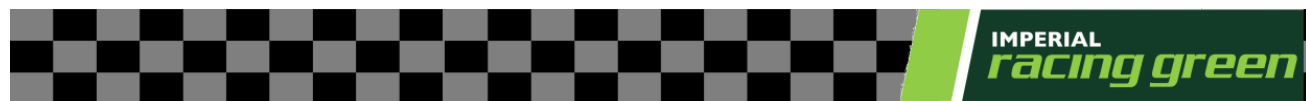
Department of Aeronautics
South Kensington Campus
Imperial College London
London SW7 2AZ
U.K.

Abstract

With a team of 18 students, the Imperial Racing Green group design project aimed to design a Class 1 vehicle, in a time span of 5 weeks, to compete in the Formula Student Event at Silverstone in July 2017, project name EV4. Over the 5 weeks, my main contribution was in the design of the suspension of the car to give optimal mechanical grip & in creating a rear wing design to produce additional downforce to improve traction & cornering capability.

The suspension was chosen to be a double unequal wishbone pullrod actuated system with a coilover spring-damper system mounted to the floor of the chassis to give the lowest possible CG position. The rear wing was optimised as a triple element wing with a drag reduction system mounted to the mainplane allowing actuation of the upper flap element to reduce drag when the car is power limited, giving optimal downforce during traction/grip limited sections and neglecting the drag penalty that such an arrangement gives at high speeds while on straight sections. The rear wing produced the largest amount of downforce of any component on the car, generating 163 N at a penalty of 57 N drag at 15 m/s.

Additional work was carried out with the vehicle performance team offering advice on possible changes to the lap simulator and aiding in the selection of tyres for EV4, designing the final livery of the car that could be used to demonstrate potential sponsorship opportunities for the business case and in producing the CAD model for the rear wing and mounting assemblies. Unlike the Rear Wing and Suspension, these contributions are not mentioned in greater detail in the report.



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Glossary

- **Camber Angle:** The angle between the vertical axis of the wheels used for steering and the vertical axis of the vehicle when viewed from the front or rear.
- **Toe Angle:** The symmetric angle that each wheel makes with the longitudinal axis of the vehicle, as a function of static geometry, kinematic and compliant effects.
- **Caster Angle:** In side-view, the inclination of the steering axis relative to the vertical.
- **Bump Travel:** The condition where the vertical load acting on a tyre is greater than the setup load leading to positive (vertical) wheel travel.
- **Droop Travel:** The condition where the vertical load acting on a tyre is less than the setup load leading to negative (vertical) wheel travel.
- **Rake Angle:** The angle formed by the front and rear ride heights of the vehicle in side view.
- **Load Transfer:** The dynamic transfer of load between the tyres the occurs due to lateral/longitudinal acceleration/deceleration. It is defined as

$$\text{Longitudinal Load Transfer (N)} = \frac{\text{Long.Acc.}(g) \times \text{CGHeight} \times \text{VehicleWeight}}{\text{Wheelbase}}$$

$$\text{Lateral Load Transfer (N)} = \frac{\text{Lat.Acc.}(g) \times \text{CGHeight} \times \text{VehicleWeight}}{\text{Wheelbase}}$$

- **Roll Centre:** The point about which the sprung mass rotates in roll.
- **Roll Couple:** The perpendicular distance between the sprung centre of mass and the roll centre
- **Motion Ratio:** The ratio of wheel displacement to spring displacement.
- **Rocker:** A type of crank that changes motion through an angle. The angle is most commonly 90 degrees or 180 degrees. Also known as a bellcrank.
- **Pullrod/Pushrod:** A rod that connects the upright to the rocker. A pull rod is only in tension while a push rod is only ever in compression.

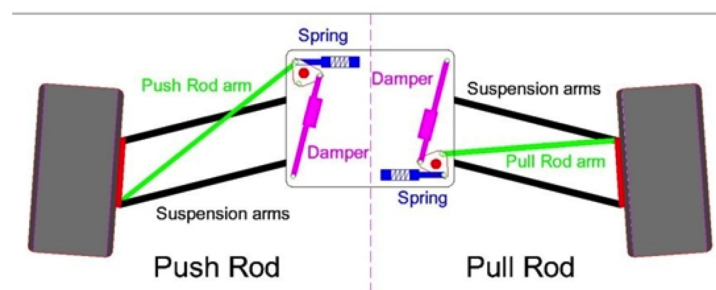


Figure 1: Example of an Inboard Pullrod and Pushrod Configuration

- **Sprung Mass:** The mass of the car supported by the suspension springs which includes the chassis frame, bodywork, engine & driver. Half the suspension arms are also considered to be sprung.
- **Unsprung Mass:** The mass of the car supported directly through the road without passing through the suspension springs, which includes the wheels and wheel assembly. Half the suspension arms are also considered to be sprung.

Suspension Design

In the Formula Student competition, vehicle apex speeds are often less than 50 KPH. At such a speed good mechanical grip is vital to provide a car that can be competitive. Mechanical grip is provided solely by the interaction between the chassis, suspension and tyres. The suspension is designed to achieve two requirements: Optimise the grip produced by the tyres by effective camber control over the course of a lap & to transmit the loads produced by the tyres into the chassis without any failure of suspension components.

1.1 Tyre Data

Tyres were selected using the Milliken tyre model to accurately predict the performance of the tyre under varying conditions. It was concluded that a 20" Goodyear tyre gave optimal grip and was chosen for the vehicle [1]. Tyre data is required to examine suspension performance as suspension geometry should be designed to maintain the grip available to the driver to close to optimum throughout the lap. After the tyre was chosen, data on the effect of camber on lateral and longitudinal grip was obtained from the tyre model:

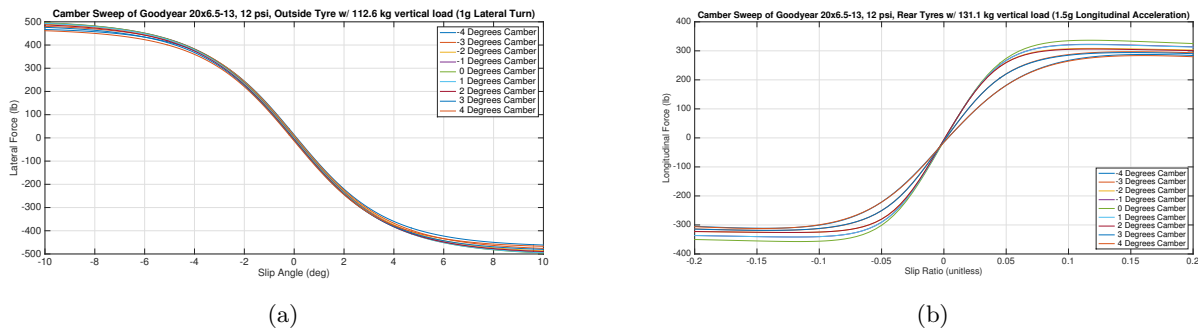


Figure 2: Tyre model outputs (a) lateral force against slip angle & (b) longitudinal force against slip ratio

On the front axle, the most important quantity to consider is the effect of camber angle on lateral force produced as this is critical during corner entry, where most weight will be on the front axle and hence a larger lateral force can be produced. From Figure 2(a) it is clear that peak grip on the outside (more heavily loaded) tyre occurs at -1° camber and it was also found the inside (less heavily loaded) tyre provides peak grip at $+1^\circ$ camber. Therefore, the front tyres will be setup at 0° camber such that when a wheel moves into bump the camber angle is negative and when the wheel is in droop the camber angle is positive. During acceleration, load is transferred towards the rear axle hence larger longitudinal forces can be produced. From Figure 2(b), Optimum traction is provided at 0° as the contact patch size is largest [2]. Hence, the rear setup camber angle was decided to be $+1^\circ$ such that at corner exit when the wheel moves into bump the camber angle is $\sim 0^\circ$.

1.2 Suspension Configuration

As camber control is of utmost importance during load transfer, double unequal length wishbones were selected allowing for varying camber throughout the lap. Such a design requires the use of either a push or pull rod to connect the spring/damper assembly to the upright, allowing the springs and dampers to be moved inboard, reducing the unsprung mass. The advantages and disadvantages of the two systems are:

- A push rod suspension is mounted higher on the chassis raising the CG position but making the components more easily accessible. This is aerodynamically more efficient if a high nose configuration is selected [3]
- A pull rod suspension allows the spring/damper assembly to be mounted to the chassis floor lowering CG position which reduces load transfer. Pull rods are generally in tension [4] and can be lighter than an equivalent push rod design.

It was determined that a pull rod configuration would be most beneficial due to the packaging and CG advantages the system offered. As a low nose configuration was chosen by the bodywork team, the aerodynamic advantages of a push rod were neglected, hence a pull rod was selected for the front/rear suspension.

1.3 Geometry Design

The suspension geometry determines the variation of camber angle with wheel travel and hence was designed such that it provides the appropriate range of wheel travel determined in Section 1.1. Another critical property is the roll centre height. This determines the amount of body roll that occurs during cornering, a property that must be controlled to provide responsive cornering and should be in the region of 0.2° - 0.7° roll/g [4]. The use of a space frame limited the positions in which the suspension could be mounted making the inclusion of anti-dive and anti-squat impossible without extensive chassis redesigns.

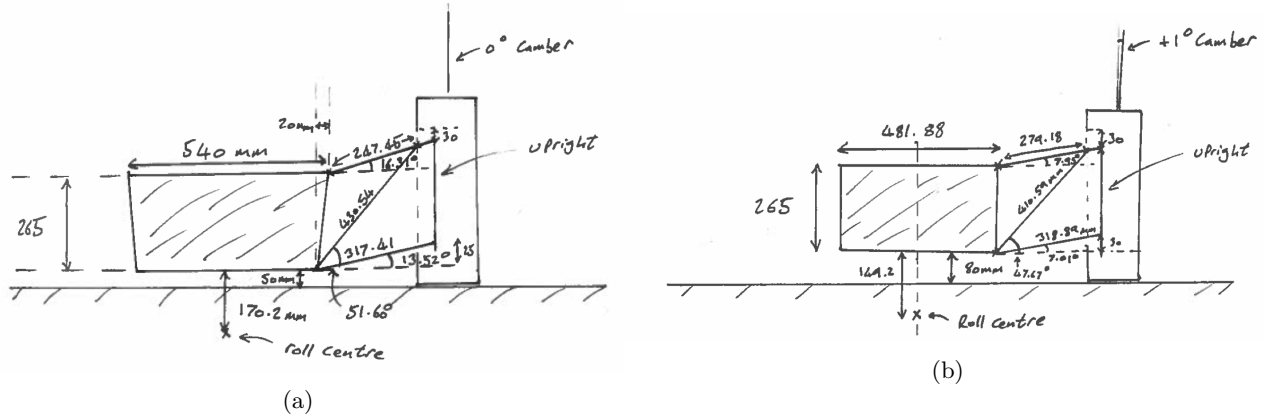


Figure 3: Dimensioned sketch of the suspension geometry at the (a) front & (b) rear of the vehicle

Dimensions of the suspension geometry are provided above in Figure 3. Both geometries produce negative camber when the wheel moves into bump giving optimal grip when the tyre becomes more loaded. The roll centre at the rear axle was 22 mm higher than at the front axle to account for the higher axle CG at the rear, primarily due to the mounting of the rear wing above the chassis floor. This allowed the roll couple to be constant throughout the car at 436.5 mm which will give the customer a car that doesn't experience jacking (where the car rises up during hard cornering [5]). The mounting points on the front upright were placed to give a $+5^{\circ}$ caster angle, providing a more sensitive steering system which responds quicker to driver inputs.

1.4 Spring & Damper Selection

With the geometry designed for the front and rear, it is possible to relate the vertical wheel travel to the movement of the pullrod mounting point. This is known as spring travel [4] as the pullrod movement dictates the spring extension/compression. These plots are presented in Appendix A. Formula Student regulations dictate that the car must have usable wheel travel of at least 50.8 mm, 25.4 mm bump and 25.4 mm droop. From Appendix A, it was possible to determine spring rates for both the front and rear suspension to meet this criteria:

Table 1: Suspension Spring Rates & Wheel Travel

Parameter	Front Suspension	Rear Suspension
Spring Rate (N/mm)	23.62	28.03
Average Motion Ratio	1.70	2.17
Wheel Rate (N/mm)	8.13	5.93
Max. Bump Travel (mm)	43	63.59
Max. Droop Travel (mm)	26.99	25.4
Setup Load (N per tyre)	695.8	849.3
Roll Stiffness (Nm/Deg.)	102.7	74.80

Table 2: Suspension Damping Rates & Coefficients

Parameter	Front Suspension	Rear Suspension
Low Frequency Bump Damping (N/m/s)	2757	2455
High Frequency Bump Damping (N/m/s)	1379	1228
Low Frequency Rebound Damping (N/m/s)	6204	5525
High Frequency Rebound Damping (N/m/s)	3102	2762
Sprung Damping Coefficient	0.7	0.7
Unsprung Damping Coefficient	0.413	0.477

Table 1 demonstrates that regulation T6.1.1 has been met with a droop travel of 26.99 mm on the front suspension and 25.4 mm on the rear suspension. Any travel greater than this will be accommodated by rising rate Eibach bump stops [6] to prevent suspension failure. With the spring rates decided, it was possible to obtain the variation of wheel travel with vertical load using the values obtained from the lap simulator:

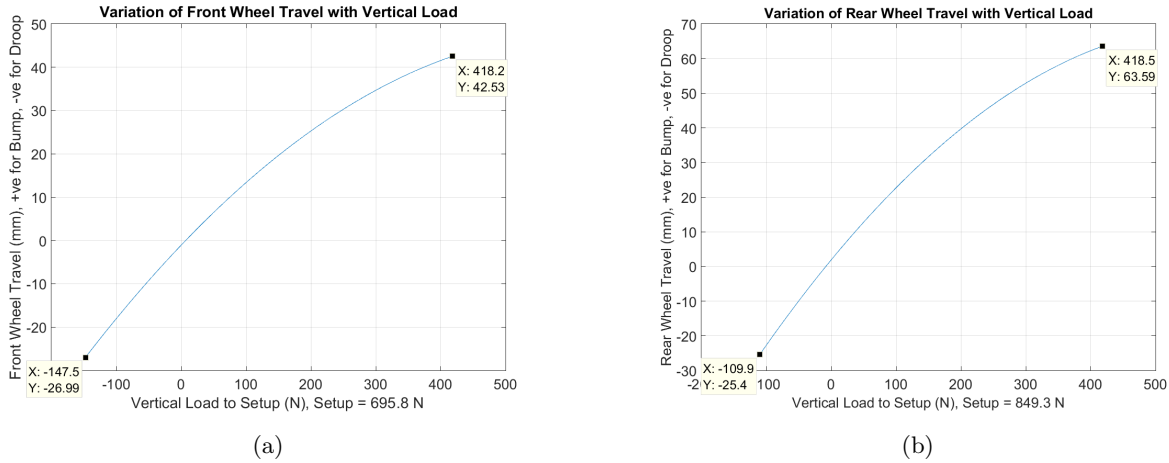


Figure 4: Wheel travel for varying vertical loads relative to setup load for (a) front suspension & (b) rear suspension

Figure 4 demonstrates that the suspension geometry chosen is a rising rate suspension in bump with the load per mm vertical wheel travel increasing as wheel travel increases. This is beneficial as it allows minimal camber changes on the outside maximum loaded tyre during the turn in phase.

Damping is critical to prevent oscillations of both the sprung and unsprung masses of the car. Racing cars often have a damping ratio in the region of 0.7 [4], higher than the average passenger car as comfort is less of a concern. For optimal damping, it was decided to choose a 4 stage monoshock damper, allowing for different damping rates in bump and rebound for high and low frequency oscillations. This is important as a wheel in bump (i.e. a kerb strike) is often travelling with much higher velocity than a wheel in rebound falling by the effect of gravity. Hence, as damping force is proportional to velocity, a lower damping rate is required in bump to give uniform damping force. Separate damping for high and low frequency oscillations is necessary to prevent large forces being transmitted during kerb strikes. Hence a wheel vertical velocity of 0.1ms^{-1} was set as the cross-over between an oscillation expected during the course of a lap and a more violent event (i.e. kerb strike). The summary of damping coefficients can be found in Table 2 with calculations presented in Appendix B.

The final spring-damper assembly was chosen to be a coilover configuration, where the spring sits around the damper. This allows tight packaging on the lower chassis member, lowering CG position. The full set of springs and dampers for the car costs £591, however such a set is standard for open wheel race cars and allows tuning of spring and damping rates during events to suit driver preference.

1.5 LapSim Analysis

In order to determine the performance of the suspension geometry selected, a wrapper was written for the LapSim code [1] to determine camber angles and ride height from the vertical tyre loads outputted by the LapSim. Figure 4 relates vertical load to wheel travel and this can be used to determine the camber angle over the course of a lap, shown above in Figure 5(a). For the front suspension the camber angle is kept between $+1^\circ$ and -2.2° throughout the course of a lap. This is in the range determined in section 1.1 & hence should give near optimal lateral tyre performance to the customer. Examining the rear suspension camber angles, at corner exit the camber angle is consistently 0° giving the optimal tyre longitudinal performance under acceleration. This is vital in Formula Student which has many slow corners.

The final aim was to provide a ride height envelope from LapSim that could give minimum clearance distances during a lap. This is essential as the setup ride height is rarely (if ever) experienced through the course of a lap due to dynamic load transfer. This envelope, Figure 5(b), was passed to the rest of the aerodynamics team to guarantee no aero surface was mounted too low under the chassis floor, avoiding any surface grounding over the course of the lap. This data also forms the basis of the aeromap presented later (Section 2.11).

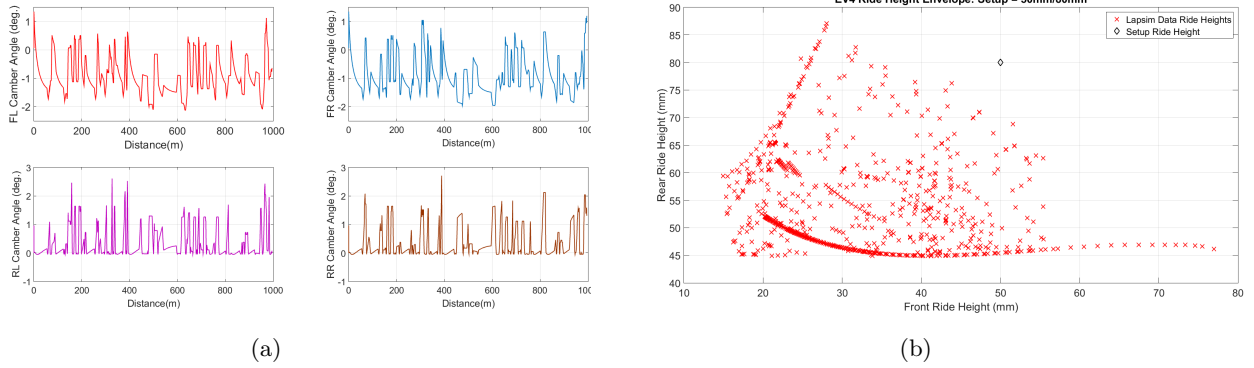


Figure 5: (a) Camber Angle Variation over the course of a lap of the Silverstone Endurance track & (b) Ride Height Envelope of the GDP 2016 Design; Setup ride height is marked as a diamond at 50/80 mm

1.6 Suspension Loads

With the geometry designed, it is necessary to meet the second objective set out and determine wishbone/ pullrod thicknesses to prevent any structural failure throughout a lap. The suspension arms were sized considering the worst case loadings sustained throughout the lap, Appendix C shows the full calculations. The minimum total assembly mass was 1.16 kg for the front suspension geometry and 1.48 kg for the rear suspension with a 50% safety factor for unforeseen collisions with kerbs/cones, giving a total suspension assembly mass for the car of 5.28 kg using 4130 Steel for the suspension arms. It was concluded that the additional cost of carbon fibre wasn't necessary considering the magnitude of internal loads the suspension arms experience at maximum corner loads. The loads were passed to the chassis team for dynamic load analysis at the suspension mounting joints.

1.7 Future Work

At present the roll stiffness of the car is unadjustable as it is determined solely by the springs selected in Section 1.4. Using the roll stiffnesses presented in Table 1, it is possible to determine:

$$\text{Roll Gradient} = \frac{\Phi_r}{A_y} = \frac{\text{Weight} \times \text{Roll Couple}}{\text{Front Roll Stiffness} + \text{Rear Roll Stiffness}} = 0.176 \text{ deg/g} \quad (1)$$

This is slightly stiffer than the optimal value. Hence, a useful addition would be to introduce an anti-roll bar system such that the roll stiffness of the suspension can be adjusted. Suspension systems should be highly tuneable to meet the preference of the drivers in the Formula Student dynamic events, this was one of the main reasons an adjustable coilover spring-damper system was selected. The main method for tuning a suspension is through track testing, which is highly recommended to the customer before the Formula Student event in 2017.

2 Rear Wing Design

One of the main directives of the 2016 GDP was to determine if it is beneficial to run an aerodynamics package in the Formula Student event. To consider this, all design decisions were incorporated into the lap simulator to determine the overall effect of downforce, drag and mass on total points scored. The values provided by the LapSim team were [1]:

Table 3: Effect of Downforce, Drag & Mass on Overall FSAE Performance

Parameter	Downforce	Drag	Mass
Points Gain	9.8 per SC_L	-30.0 per SC_D	-0.7 per kg

The rear wing forms a vital part of any aerodynamic package, balancing the downforce produced by the front wing to give a centre of pressure position that optimises cornering potential & producing a region of low pressure

near the diffuser exit that improves undertray performance. However, as the most rearwards component of the car it also receives little uniform, laminar airflow & this had to be considered in the design.

Due to the large differences in reference area (S) that can occur in the design of a large component such as a rear wing, dimensional lift and drag coefficients, SC_L & SC_D (Units: m^2) are quoted in this report. This allows direct comparison between designs, simply multiplying by the dynamic pressure, $\frac{1}{2}\rho v^2$, to obtain the downforce and drag produced.

2.1 CFD Validation

Star-CCM+ was one of the main components used in making design decisions during this project and as such initial simulations compared the physics models in Star-CCM+ to determine an efficient & robust platform on which CFD simulations could be run. The models chosen were:

- **Steady Flow:** While many vortices are shed upstream of the rear wing & the Reynolds number of the flow the car experiences almost certainly means these are turbulent unsteady vortices, a steady flow model was chosen in order to increase the number of concepts that could be evaluated due to reduced simulation times.
- **Segregated Flow:** A comparison of convergence times was carried out and it was determined that segregated flow simulations converged 40% faster than the coupled flow simulations. The main advantage of solving coupled equations for velocity and pressure is in capturing large density fluctuations [7], however as our car operates in the inviscid region ($M < 0.3$) this shouldn't be a concern.

A polyhedral mesh was chosen due to the faster convergence it offered compared to a tetrahedral or hexahedral mesh. Initial simulations with tetrahedral simulations suffered from fluctuating residuals that failed to converge however sims run on a polyhedral mesh were more robust and hence were chosen for the final simulation platform.

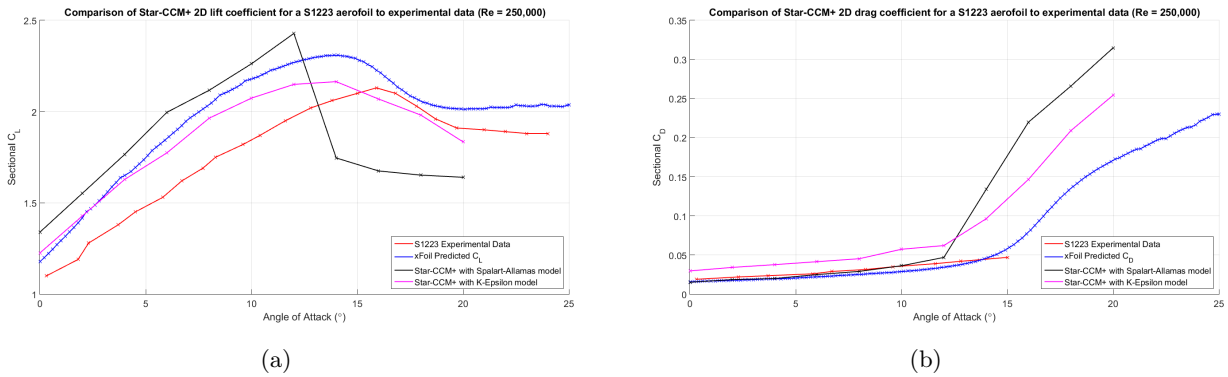


Figure 6: Comparison between 2D Star-CCM+ simulations and experimental data for (a) Lift coefficient & (b) Drag coefficient

Two-dimensional CFD simulations were run in Star-CCM+ and comparisons were made to xFoil and experimental data obtained from the University of Illinois [8] to estimate any errors that may be present in the prediction of lift and drag in Star-CCM+. The study, shown in Figure 6, showed that at low angles of attack both the Spalart-Allmaras & K- ϵ model overestimated the lift coefficient by 20% with the K- ϵ model overestimating drag by as much as 60%. However near separation the K- ϵ model accurately predicted C_L within 5% but both turbulence models overpredicted drag near separation by as much as 120%. As the K- ϵ model more accurately captured downforce levels near separation & more accurately predicted the angle at which separation occurred, it was concluded this model would be more suitable for the wing optimisation study. However, caution should be taken with the drag values predicted in this report as they are likely overestimates especially when the elements are near to separating.

2.2 CFD Setup

A Star-CCM+ simulation platform was created to allow direct comparisons to be made between different rear wing designs. The control volume used, shown in Figure 7(a), had an upstream length of 3 car lengths to allow

the flow to develop prior to reaching the model. A downstream length of 5 car lengths and a domain height of 4 car lengths were selected allow the wake to develop behind the model.

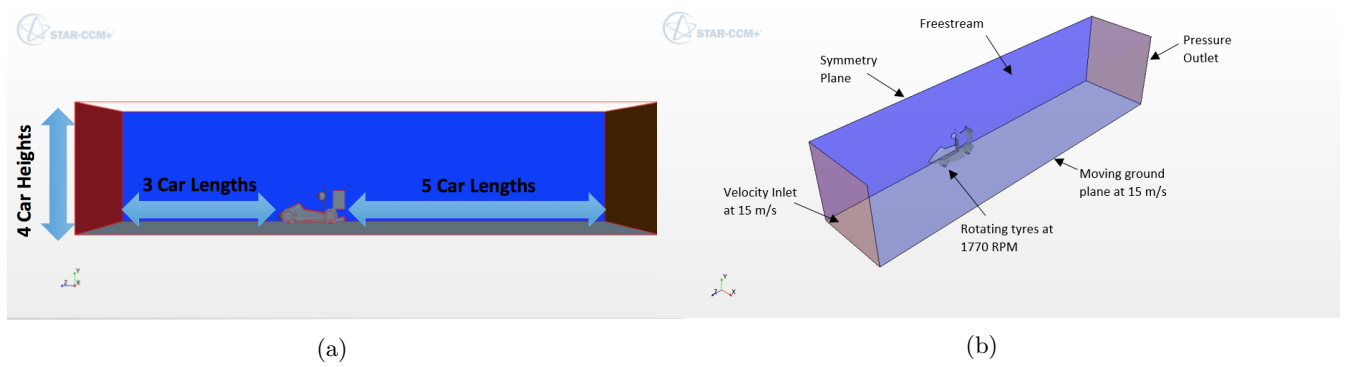


Figure 7: Star-CCM+ Control Volume setup detailing (a) region dimensions & (b) boundary conditions applied

Boundary conditions (Figure 7(b)) were set to best replicate the conditions the car would experience on track. Initial simulations were run as a straight line test at 15 m/s (the average velocity of the car throughout a lap) with a moving ground plane at 15 m/s added in order to give the best approximation of the car's ground effect, which is crucial to maximising an open wheel race car's downforce. The rear wing will also be greatly affected by the separated flow shed from the roll hoop and chassis which is immediately upstream of the wing. Hence all models contained the rear wing to be tested, along with the chassis, roll hoop and driver's head. Initial comparisons suggested these upstream effects lead to a 20% loss of downforce (Appendix D), mainly due to lost downforce at the centre of the wing, which sits in the wake of the driver's seat. Wing simulations optimising endplate design also considered the effect of rotating tyres as the drag produced in the interaction between the endplate and tyre wake is critical in determining the overall efficiency of the wing.

2.3 Mesh Convergence

Prior to simulations being run, a mesh convergence study was carried out to optimise the number of cells required to achieve maximum computational efficiency. Volumetric control was employed with additional refinement for the rear wing & chassis to accurately mesh the geometry being considered.

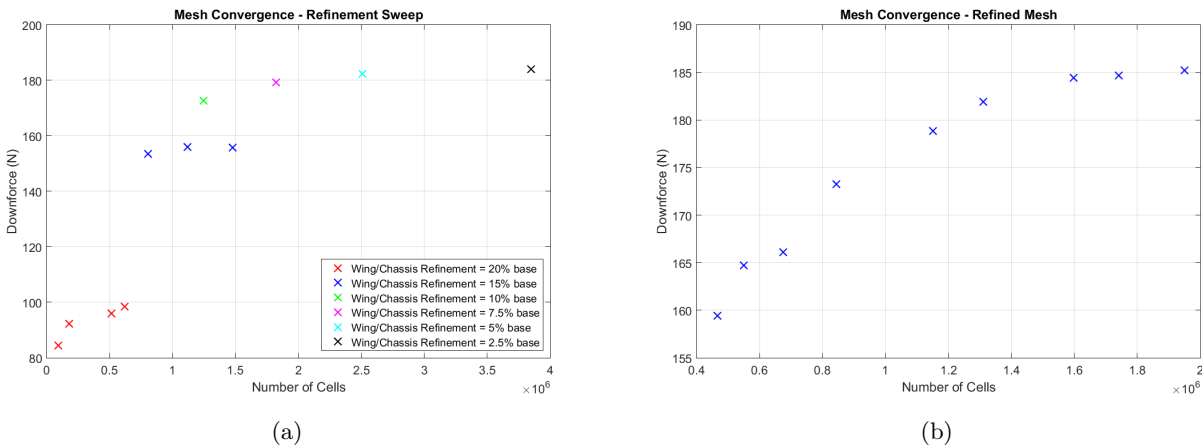


Figure 8: Mesh Convergence Sweep for Rear Wing Simulations

The initial convergence studies focussed on determining the maximum refinement percentage to base size necessary to accurately capture the flow pattern of the geometry. Figure 8(a) shows that a refinement of 5% base size was sufficient to show convergence in downforce levels estimated in CFD simulations.

A second sweep was carried out with the volumetric control cylinders added at the wing element leading edges. The refinement region around the wing and chassis was set to 5% base size and the base size of the control volume was varied until convergence was reached. Figure 8(b) demonstrates downforce is approximately constant after 1.6 million cells were used & hence this was considered to be the minimum number of cells necessary to give accurate, converged results. These values were employed in all other simulations throughout the design process.

An additional check was carried out by varying inlet flow velocity for a mesh of 1.6 million cells. As expected, drag and downforce both showed approximately a quadratic relation with flow velocity (Appendix E).

2.4 Rear Wing Regulations

Aerodynamic devices are stringently regulated by FSAE & failure to comply with the regulations will lead to disqualification from the event. The regulations relating to the rear wing are [9]:

- T9.3.1 In plan view, no part of any aerodynamic device can be: a) Further rearward than 250 mm rearward of the rear of the rear tires & b) Wider than the inside of the rear tires.
- T9.3.2: In side elevation, no part of the rear wing or aerodynamic device (including end-plates) may be higher than 1.2 meters above the ground when measured without a driver in the vehicle.

Appendix F shows a visualisation of these regulations. Based on this envelope it was decided that the rear wing would be mounted at the maximum height to reduce the effect of the tyre wakes and chassis wake (the chassis narrows with vertical height), optimising wing performance. The rearmost surface of the wing was placed at the maximum 250mm behind the rear tyres to reduce the effect of the roll hoop vortices on the wing, increasing the amount of laminar air reaching the wing. Due to the low maximum speed a large reference area is needed to generate sizeable amounts of downforce hence the maximum allowable width was selected.

2.5 Aerofoil Selection

In order to determine the most suitable aerofoil choice for the rear wing mainplane and flaps, an initial comparison of aerofoils designed for $Re = 500,000$ flow was carried out using xFoil. The data obtained from this comparison is presented in Appendix G. The mainplane aerofoil was chosen to be an S1223 aerofoil as it gave the highest peak C_L (a maximum of 2.29 at 12.6° , 0.23 greater than any other aerofoil tested). The flaps were manufactured using a DAE-11 aerofoil, which gives less peak downforce than the S1223 but has reduced downforce drop-off after stall which is particularly important for flaps which will operate very close to stall, especially when pitching of the car under braking is considered. Additionally, the DAE-11 is a thicker aerofoil which will make the smaller chord flaps easier to manufacture [10].

2.6 Initial Concepts & Hypothesis

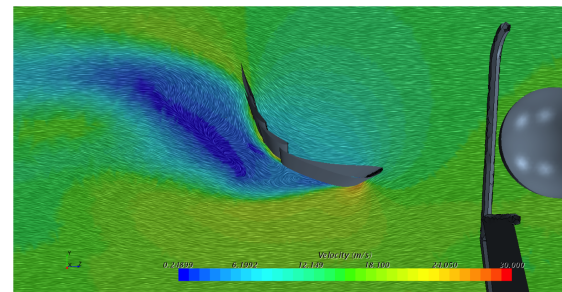
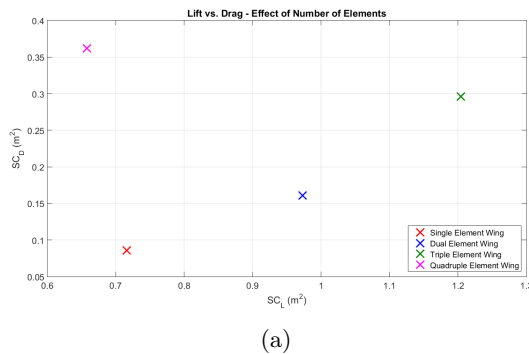


Figure 9: (a) Results of CFD simulations of initial sweep of single and multi element wing designs & (b) CFD vector visualisation of wake region and mainplane separation for quadruple element wing

The initial point estimates provided by LapSim suggested a minimum L/D ratio of ~ 3 is required to gain points. As the largest aerodynamic device, the rear wing often generated large amounts of drag on straights. In

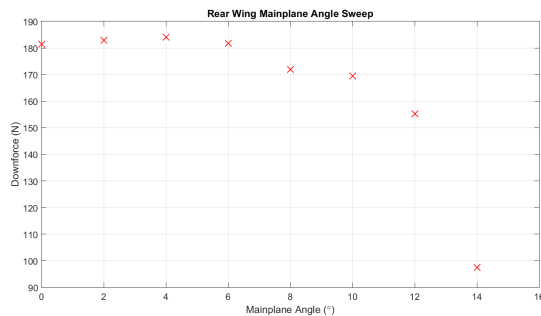
these sections the car is said to be 'power limited' & hence additional downforce isn't necessary. However, the downforce is beneficial in regions where the car is 'grip limited' meaning the car is on the performance of the tyre. It was decided that in order to increase apex speeds without the drag penalty of running a high downforce configuration in straights, a drag reduction system (DRS) should be implemented to actuate the flaps to minimise drag during straights. Hence when choosing flap angles the only objective was to maximise downforce as the design configuration will only be experienced in grip limited regions where drag is of secondary importance.

An initial sweep of wings of elements ranging from a single element to a quadruple element wing was carried out to determine the number of elements the wing should consist of. The results can be seen in Figure 9(a), the downforce produced by the rear wing increased up to the triple element wing (which generated 166N downforce at 15 m/s, 40% more than the single element wing). However, there was a 45% decrease in downforce when a fourth element was added. Figure 9(b) shows the velocity field around the wing and it is clear that the height of the quadruple element wing leads to a large wake forming behind the wing (similar to the flow field that would be expected with a bluff body) causing mainplane separation and a loss in downforce generated. It was concluded that no more than 3 elements should be considered for the rear wing design and as maximum downforce is generated with a triple element wing this configuration was selected for further optimisation.

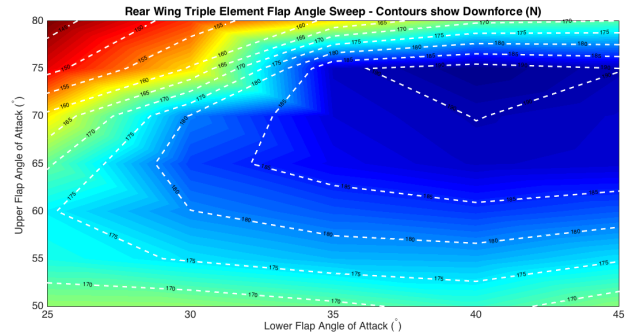
2.7 Parameter Sweeps

2.7.1 Mainplane Angle

The first optimisation centred around the angle of the mainplane. This parameter is greatly affected by the flap angles which modify the flow around the mainplane compared to a freestream simulation. Hence flap angles & flap to mainplane chord ratio were initially selected from ESDU 94031 [11], which features sets of windtunnel data for triple element wings. The CFD scan results are shown in Figure 10(a) and revealed that mainplane separation occurred at 8° , much lower than the 2D aerofoil simulations run for validation. Maximum downforce of 184N was generated at 4° and is roughly constant from 0° to 6° . Hence with a mainplane angle of 4° there should be little variation in downforce produced due to the vehicle pitching (which causes an rear wing angle of attack variation of $\pm 2.25^\circ$) and so this was fixed as the mainplane angle.



(a)



(b)

Figure 10: (a) Graph summarising downforce produced by mainplane angle sweep CFD simulations & (b) Results of Flap Angle Sweep. Blue shows higher downforce configurations, red shows lowest downforce

2.7.2 Flap Angles

Similar to the mainplane, the interaction between the two flap angles is critical in determining if separation will occur. Hence a two parameter simultaneous sweep of flap angles was carried out to determine the maximum downforce configuration. Figure 10(b) shows a 5% downforce increase to 192N was obtained with a lower flap angle of 40° and an upper flap angle of 75° . However, this places the upper flap close to separation (a 5N per degree decrease in downforce can be seen when upper flap angle increases above 75°). Hence, a more conservative upper flap angle of 70° was selected, reducing downforce produced to 190N but making the car less prone to sizeable variations in downforce under heavy braking and during corner entry where stability is key.

2.7.3 Slot Gap

The slot gap height is the vertical height between elements in the wing. A sweep of heights ranging from 5mm to 20mm was carried out in 5mm steps & it was concluded that maximum downforce was generated at 15mm producing 191N. However, downforce only varied by 10N across the range of slot gap heights tested and hence no additional tests into the slot gap height parameter was carried out. Literature suggests the position of maximum thickness of the flap should align with the trailing edge of the element below it to accelerate the flow through the slot gap and reduce the chance of separation on upper elements [10]. As this is considered to be the optimum horizontal positioning of the flaps, a sweep of overlap distance was deemed unnecessary in the restricted timeframe of this project (however this could be carried out in future to evaluate if any performance gains are possible). The results of the slot gap sweep are plotted in Appendix H.

2.7.4 Gurney Flap

By fitting a short strip perpendicular to the trailing-edge of a wing it is possible to increase both the CL generated and the maximum CL, with little increase in zero-lift drag [12]. This is known as a Gurney flap and is a highly efficient way to generate additional downforce. A Gurney flap was added to the uppermost flap with a height of 0.05c (15mm) and was tested in Star-CCM+. This gave a downforce increase of 10N to 201N of downforce generated with a drag increase of only 2.5N at an efficiency ratio of 4:1. Hence this was considered to be beneficial to the overall rear wing performance and was added to the final wing design.

2.7.5 Beam Wing

A beam wing is the concept of placing a second mainplane at the bottom of the endplates to provide a lower pressure area at the exit of the diffuser to generate more undertray downforce. The wing itself will also generate additional downforce, however this occurs at a reasonable drag penalty. An initial concept was evaluated in Star-CCM+ to determine if the concept should be further evaluated. The simulation predicted a 14N increase in downforce for the wing and diffuser combined with a drag penalty of 6.7N. As the beam wing cannot be actuated to reduce drag during straights, this efficiency gain isn't enough to yield a net points increase. Also, due to the additional complications of manufacturing a wing with a beam wing, it was concluded a beam wing wasn't beneficial to this design without extensive optimisation and hence no further studies were carried out.

2.8 Endplate Design

Endplate design is critical in optimising a rear wing's efficiency, they play a major role in reducing 'spillage' helping to maintain the pressure gradient across the upper and lower surface of the wing. They also can move the effect of vortices away from the wing surface decreasing their detrimental effect on performance [10].

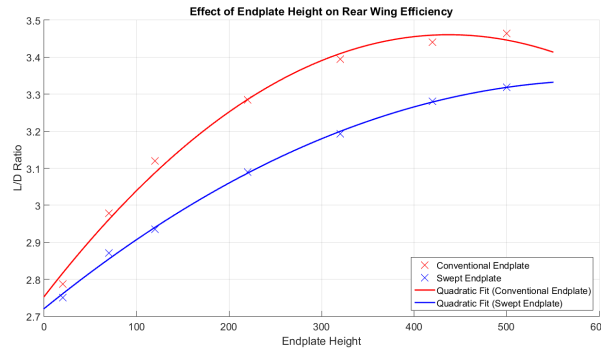


Figure 11: Variation of wing efficiency with rear wing endplate height

An initial study was carried out between a conventional rectangular endplate and a swept endplate (similar to those to be employed in 2017 F1 cars [13]) to determine which concept would yield maximum efficiency. It was discovered that while both endplates yielded a comparable drag value (consistently within 1N of each other), the conventional endplate wings produced downforce levels 5-7% higher than the swept endplate designs. Hence

initial designs centred around a rectangular endplate. The sweep revealed the efficiency of the wing increased until the endplate extended 500mm below the suction surface of the wing. At this stage efficiency gains became negligible and to reduce mass, a 420mm endplate height was selected to be the optimal choice.

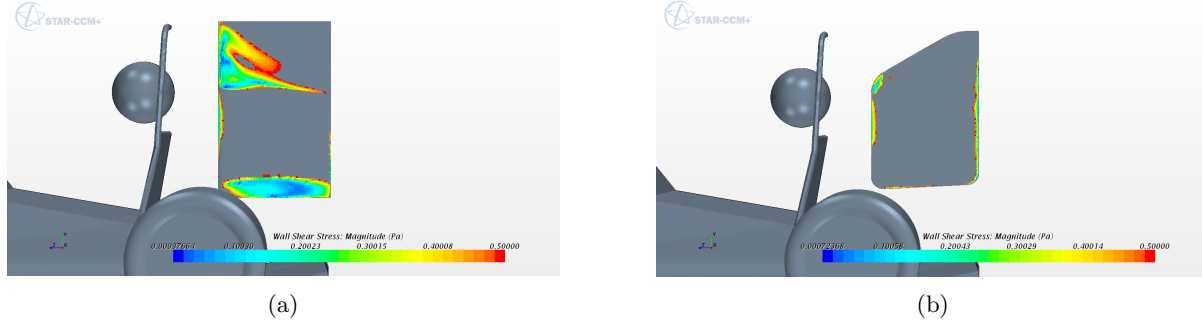


Figure 12: Regions of near separated or separated flow ($< 5\%$ max. wall shear stress) on the (a) original endplate design & (b) optimised endplate design

When considering endplate design, the interaction between the rear tyre wakes and endplates is crucial to optimising performance. Comparing the simulation with rotating tyres to those with tyre wakes neglected showed a 7.5% increase in drag. Hence additional work was carried out to reduce the drag produced by the endplates. Figure 12(a) shows the regions of separated or near separated ($< 5\%$ maximum wall shear stress) on the original endplate design and demonstrates the large wake formed by the tyres leads to a large region of separated flow on the endplate outside wall. Figure 12(b) shows the improved design which employed a lip on an angled endplate lower surface to force the tyre wake underneath the endplate and prevent any large separation regions forming. The uppermost surface was also angled to remove the large separation region present. This led to a drag reduction of 3% with the lip also producing a 3.3% increase in downforce, generating 188N at 15 m/s

2.9 Drag Reduction System

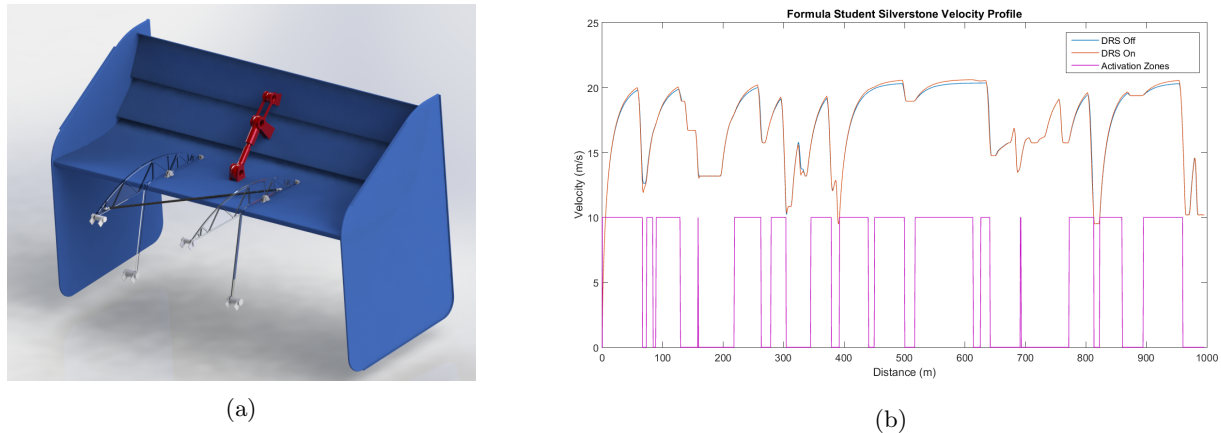


Figure 13: (a) Rear Wing CAD Model showing the position of the DRS in red & (b) LapSim plot showing possible activation zones and effect of DRS activation on car velocity

As previously discussed, the drag reduction system allows the actuation of the upper element angles of attack to reduce the drag during power limited sectors. An analysis of the Silverstone Formula Student track identified 11 sections of the track where the car is power limited and DRS could be activated (shown in pink in Figure 13(b)), ignoring regions too small for the driver to be able to activate the system. The DRS can be seen in Figure 13(a), the system works using a pneumatic actuator, weighing 0.48kg, mounted on the upper surface of the mainplane. The actuator sits in the wake region behind the driver's seat and roll hoop and hence has very

little effect on drag making the design of a shroud unnecessary. The system would be activated via a button press on the steering wheel and is closed by application of the brake pedal.

Simulations determined that with the flap angles reduced to 0° there was an 89% reduction in drag produced by the wing accompanied by a 75% decrease in downforce generated. This was highly beneficial to overall performance when accounted for in LapSim with a 3-4KPH speed gain on the longest straights on the circuit (clearly visible by comparing the velocity traces in Figure 13(b)). Overall the DRS led to a 9.1 point gain over the course of the event and is seen as critical to making the rear wing a viable component for EV4. The below table summarises the numerous iterations of the rear wing design carried out through the project:

Table 4: Rear Wing Simulation Results at 15m/s, 0° Yaw Angle

Design	Downforce (N)	SC_L (m^2)	Drag (N)	SC_D (m^2)	L/D
Initial Triple Element	165.9	1.204	43.7	0.317	4.12
Mainplane Angle = 4°	184.1	1.335	44.1	0.320	4.18
Flap Angles = $40^\circ/70^\circ$	192.1	1.394	46.7	0.339	4.11
with 0.05c Gurney Flap	201.2	1.460	49.2	0.357	4.09
0.42m Endplate and Rotating Tyres	182.2	1.332	52.9	0.384	3.44
with Optimised Endplate	188.2	1.366	51.2	0.372	3.67
DRS Enabled	46.86	0.340	5.84	0.042	8.03

2.10 Rear Wing Mountings & Structural Analysis

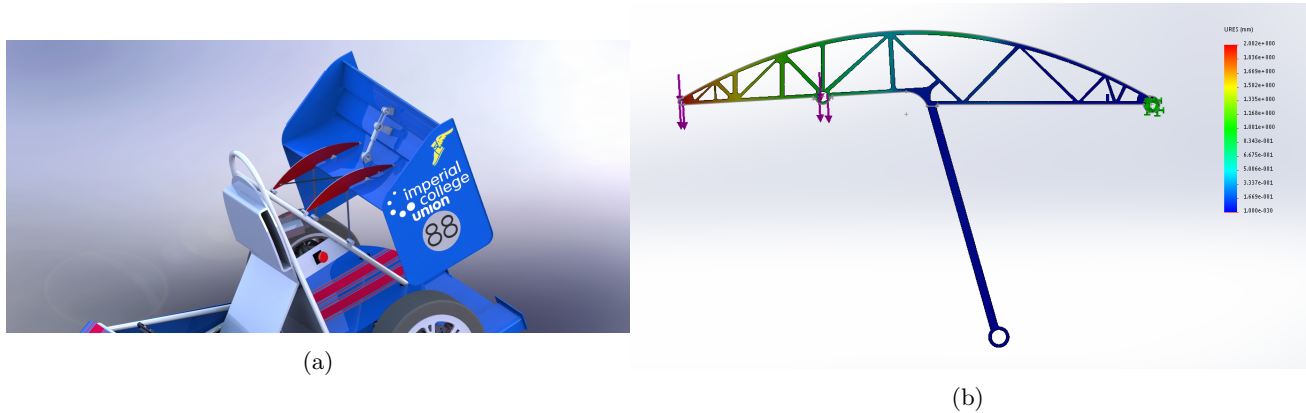


Figure 14: (a) CAD model showing the positioning of the rear wing mounting structure & (b) Finite element analysis of the swanneck structure showing 2mm maximum deflection at maximum aerodynamic load

When considering how to mount the rear wing, two concepts were initially considered: 1) The mounting points are placed on the suction surface of the wing with the wing mounting to the upright making the rear wing an unsprung mass. This would mean no angle change when the car pitches. 2) The mounting points are placed on the pressure surface of the wing with a swanneck structure connecting the rear wing to the roll hoop support bars. Option 1 was rejected as the car has a large amount of unsprung mass (20.23kg per corner) due to the in-wheel motors, which is bad for vehicle dynamics. The swanneck structure was selected as it allowed the mounting to be placed on the pressure surface of the wing reducing the efficiency loss caused by adding mounting points and also as it allowed the entire structure to be placed in the wake of the roll hoop and driver, greatly reducing the drag caused by the mounting structure. These mountings can be seen in Figure 14(a).

The wing is to be manufactured using a sandwich construction of 2mm fibreglass laminate skins with an aluminium honeycomb core to stabilize the outer skin [14]. A composite of E-glass in a polyester matrix was chosen over carbon fibre as the aerodynamic forces experienced by the wing are low even at a max speed of 25m/s and hence the additional specific strength of CFRPs is not required. Also a high quality fibreglass composite is cheaper than CFRPs (£21/kg vs. £38/kg), reducing the overall material costs. Initial structures calculations considered a $[0/0/\pm 45^\circ/\pm 90^\circ]_s$ laminate to give a symmetric, balanced laminate that could resist

shear forces on top of lateral and longitudinal forces produced. Deflection calculations for FS regulations centred about treating the wing as a cantilever with a 200N force applied at the furthest distance from the mountings. Using composite theory, it was calculated that the wing would experience a vertical deflection of 14mm with the swanneck structures experiencing a 4.2 mm deflection giving a total deflection of 18.2mm, 30% lower than the allowed deflection of 25mm.

The swanneck structure was optimised to minimise weight while still allowing minimal deflection. The end of straight deflection was calculated by FEA to be only 2mm (Figure 14(b)) with the structure weighing only 0.41kg. The design also features a carbon rod acting as a stiffener to prevent the wing translating laterally while the car is yawing. The total assembly mass was calculated to be 8.2kg (6.6kg fibreglass/polyester, 1.1kg aluminium honeycomb, 0.5kg for DRS actuator) with a material cost of £172 & the DRS actuator cost of £35. It is difficult to consider the total cost as many items would need to be purchased such as resin, foam moulds, vacuum bags etc. however other competitors suggest a total cost in the region of £1900 [15] for a full car aero package. A likely method for manufacture would be an open mold manufacturing process.

2.11 Performance Analysis

In order to obtain a full appraisal of the performance of the aerodynamic package, a full car simulation was run to determine the effect of the aerodynamic package throughout a lap. As the car is a dynamic system, it was decided to take the ride height envelope produced during the suspension design and run simulations at numerous front and rear ride height values to produce an aeromap that can be used to evaluate the downforce and drag produced by the car under any dynamic wheel loading.

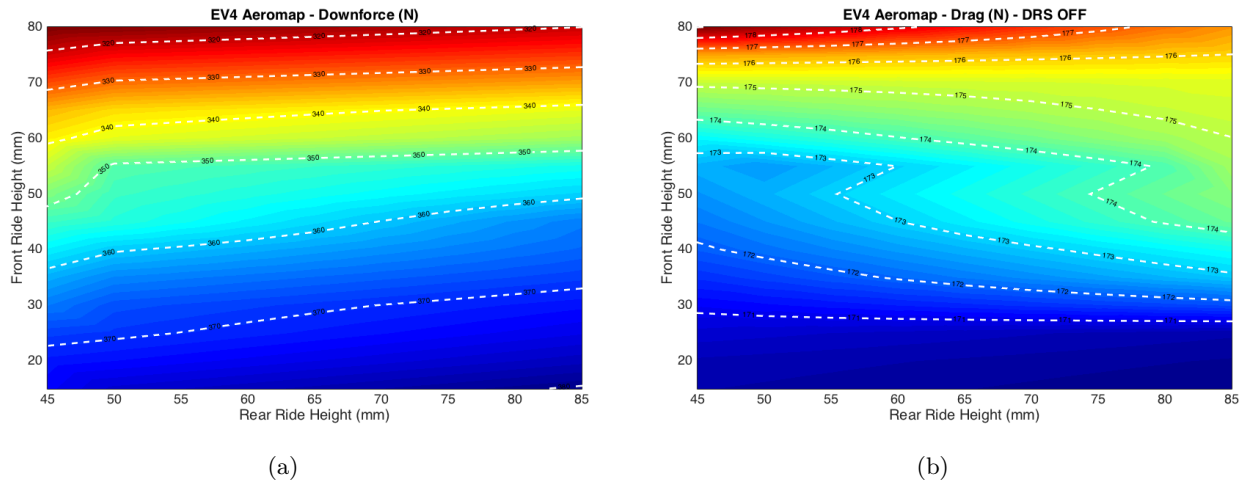


Figure 15: Final aeromap for EV4 showing the effect of ride height on (a) downforce & (b) drag

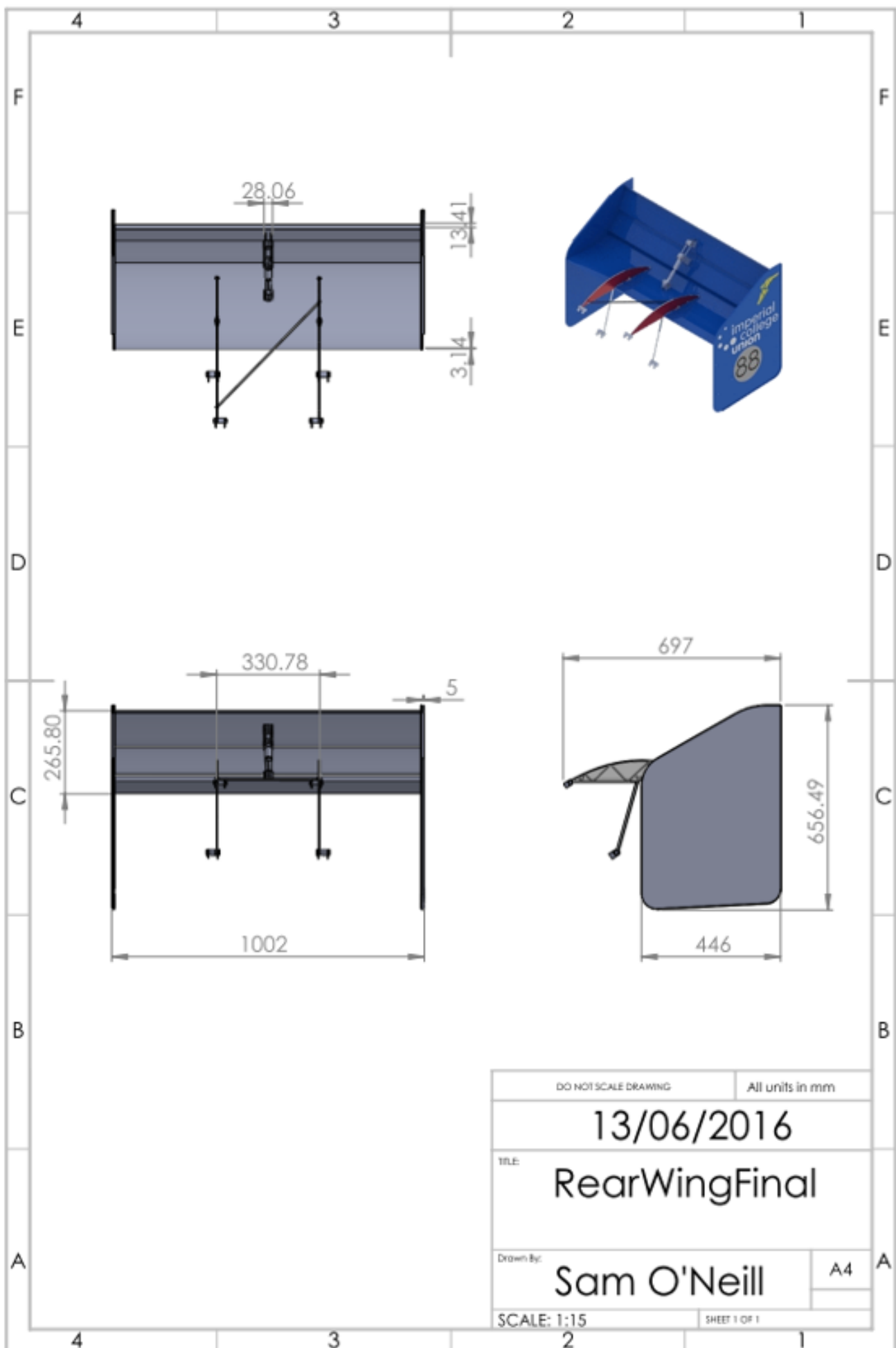
The aeromap, Figure 15, showed that maximum downforce is produced at corner entry, when front ride height (FRH) is lowest and rear ride height (RRH) at it's highest, with 381N downforce generated with 170N of drag. There is a loss of downforce however at corner exit (high FRH, low RRH) with downforce reduced to 315N and drag increasing to 178N with most of the loss of downforce coming from the front wing and diffuser. In the full car simulation, the rear wing produced 163N downforce at a penalty of 57N drag at 15m/s.

Finally, the full car model was run at a 5° yaw angle in order to determine the loss of downforce that will occur during heavy cornering. The rear wing demonstrated only a 4.8N loss of downforce, however drag increased by 12% to 56.3N at 15m/s. While the small loss of downforce is acceptable and wouldn't greatly unsettle the car, the increased drag will reduce yaw rate and introduce additional understeer to the car. Future work would focus on the addition of slots in the endplate to allow the flow to pass through, reducing the static pressure on the outside of the endplate and therefore drag.

Working with the LapSim team, it was concluded that the aerodynamic package gained 13.3 points over the course of the FS dynamic events [1]. However this comes with a high cost of £1900 and long manufacturing times. Hence the aerodynamics team suggest that the final decision be left to the customer, Imperial Racing Green, to determine which configuration they would prefer taking cost & time availability into consideration.

References

- [1] Dhall, K., Vehicle Performance: Final Report, Imperial College; 2016
- [2] Pacjeka, H., Tire and Vehicle Dynamics, 3rd Edition, Butterworth-Heinemann; 2012
- [3] Staniforth A., Competition Car Suspension: A Practical Handbook, 4th Revised Edition, J H Haynes & Co Ltd; 2006
- [4] Seward, D., Race Car Design, 1st Edition, Palgrave Macmillan; 2012
- [5] Milliken W.F., Milliken D.L., Race Car Vehicle Dynamics, Premiere Series, EDS Publications Ltd; 1995
- [6] Eibach Bump Stops Part Nr: 770002 [Unknown], Available from: <http://eibach.com/america/en/motorsport/products/accessories/bumpstops/770002> [Accessed 10th June 2016].
- [7] Bakker, A., Applied Computational Fluid Dynamics Solution Methods, Dartmouth College, Available from: <http://www.bakker.org/dartmouth06/engs150/05-solv.pdf>; 2006
- [8] Selig, M.S., Guglielmo, J.J., Broeren, A.P., Giguere P., Summary of Low-Speed Airfoil Data, Volume 1, University of Illinois; 1995
- [9] SAE International, 2015 Formula SAE Regulations, Available from: <http://formulastudent.imeche.org/docs/default-source/default-document-library/2016-fsae-rules.pdf?sfvrsn=0>; [Accessed 10th June 2016].
- [10] McBeath, S., Competition Car Aerodynamics: A Practical Handbook, 1st Edition, J H Haynes & Co Ltd; 2006
- [11] Garner, H.C. et al, ESDU 94031: Increments in aerofoil lift coefficient at zero angle of attack and in maximum lift coefficient due to deployment of a double-slotted or triple-slotted trailing-edge flap, with or without a leading-edge high-lift device, at low speeds, ESDU; 1995
- [12] Jeffrey, D., Some Aspects of the Aerodynamics of Gurney Flaps on a Double-Element Wing, Journal of Fluids Engineering, Vol. 123; 2001
- [13] Piola, G., Tech analysis: What F1 2017 car rules will actually look like, Available from: <http://www.motorsport.com/f1/news/tech-analysis-what-f1-2017-car-rules-will-actually-look-like-675492/>; [Accessed 11th June 2016].
- [14] Megson, T.H.G, Aircraft Structures for Engineering Students, 5th Edition, Butterworth-Heinemann; 2012
- [15] Anderson, M., University of Victoria FSAE Team Aerodynamic Package Development, University of Victoria, Available from: <https://www.indiegogo.com/projects/uvic-fsae-team-aerodynamic-package-development/>; [Accessed 8th June 2016].



3 Appendices

3.1 Appendix A: Wheel Travel Plots

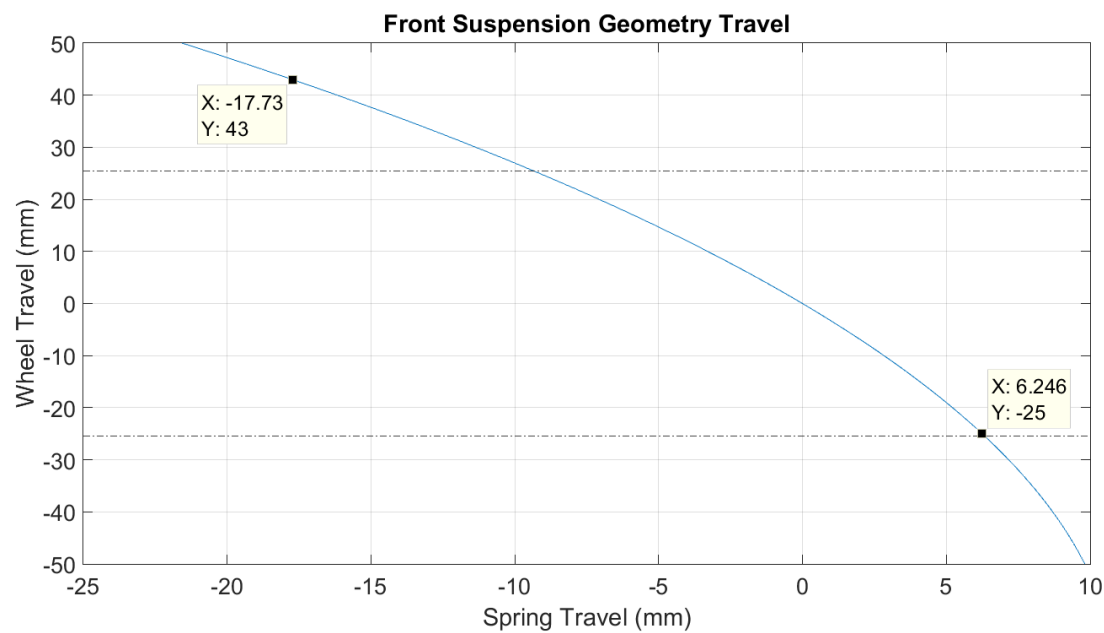


Figure 16: Suspension travels for front suspension geometry

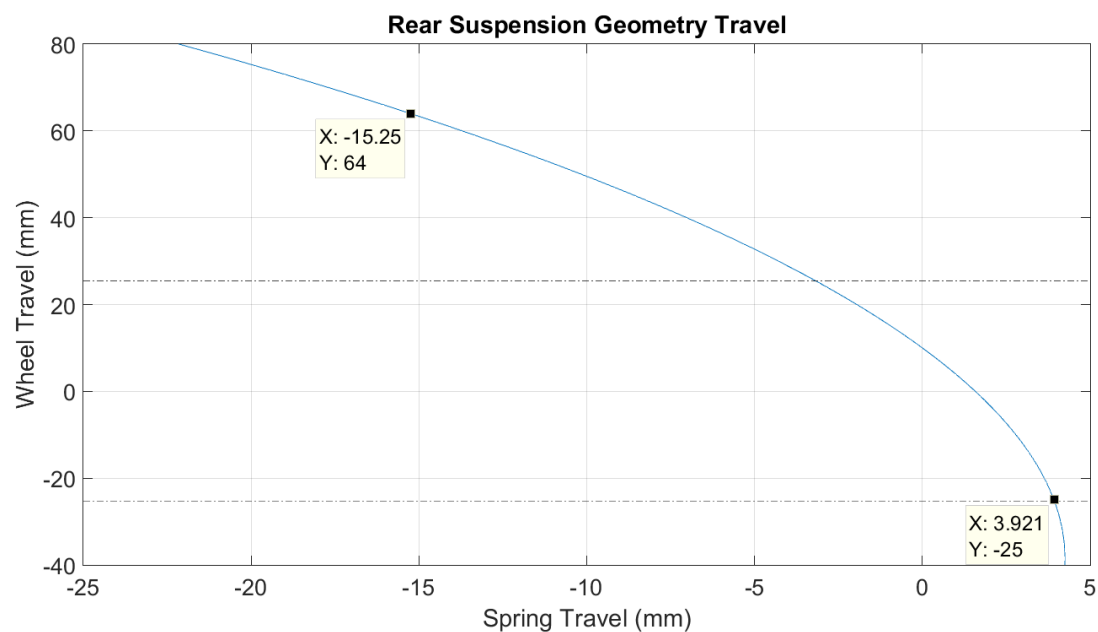


Figure 17: Suspension travels for front suspension geometry

3.2 Appendix B: Damping Calculations

Percentage damping values relative to the selected damping coefficients for low/high frequency bump/rebound damping rates are obtained from [4].

3.2.1 Front Axle Damping

$$\text{Motion Ratio, } R_m = 1.70$$

$$\text{Spring Rate} = 23.62 \text{ N/mm}$$

$$\text{Tyre Spring Rate} = 128.7 \text{ N/mm (From Tyre Data)}$$

$$\frac{1}{\text{Ride Rate}} = \frac{1}{\text{Spring Rate}} + \frac{1}{\text{Tyre Spring Rate}} = \frac{1}{23.62} + \frac{1}{128.7}$$

$$\therefore \text{Ride Rate} = 19.96 \text{ N/mm}$$

$$\text{Sprung Corner Mass} = 52.31 \text{ kg; } \text{Unsprung Corner Mass} = 20.23 \text{ kg}$$

$$\text{Sprung Natural Frequency, } f_s = \frac{1}{2\pi} \sqrt{\frac{19.96 \times 10^3}{52.31}} = 3.11 \text{ Hz}$$

$$\text{Unsprung Natural Frequency, } f_u = \frac{1}{2\pi} \sqrt{\frac{23.62 \times 10^3 + 128.7 \times 10^3}{20.23}} = 13.81 \text{ Hz}$$

$$\text{Sprung Critical Damping, } C_{crit} = 4\pi m_s f_s R_m = 5908.2 \text{ N/m/s}$$

$$\text{Selected Damping Coefficient, } C = 0.7 C_{crit} = 4136 \text{ N/m/s (0 d.p.)}$$

$$\text{Unsprung Critical Damping, } C_{crit} = 4\pi m_u f_u R_m = 10011 \text{ N/m/s}$$

$$\text{Unsprung Damping Coefficient} = \frac{C}{C_{crit}} = \frac{4136}{10011} = 0.413 \text{ (3 d.p.)}$$

$$\text{Low Frequency Bump Damping} = \frac{2}{3} C = 2757 \text{ N/m/s (0 d.p.)}$$

$$\text{Low Frequency Rebound Damping} = \frac{3}{2} C = 6204 \text{ N/m/s (0 d.p.)}$$

$$\text{High Frequency Bump Damping} = 0.5 \times 2757 = 1379 \text{ N/m/s (0 d.p.)}$$

$$\text{High Frequency Rebound Damping} = 0.5 \times 6204 = 3102 \text{ N/m/s (0 d.p.)}$$

3.2.2 Rear Axle Damping

$$\text{Motion Ratio, } R_m = 2.17$$

$$\text{Spring Rate} = 28.03 \text{ N/mm}$$

$$\text{Tyre Spring Rate} = 128.7 \text{ N/mm (From Tyre Data)}$$

$$\frac{1}{\text{Ride Rate}} = \frac{1}{\text{Spring Rate}} + \frac{1}{\text{Tyre Spring Rate}} = \frac{1}{28.03} + \frac{1}{128.7}$$

$$\therefore \text{Ride Rate} = 23.02 \text{ N/mm}$$

$$\text{Sprung Corner Mass} = 63.68 \text{ kg; } \text{Unsprung Corner Mass} = 20.23 \text{ kg}$$

$$\text{Sprung Natural Frequency, } f_s = \frac{1}{2\pi} \sqrt{\frac{23.02 \times 10^3}{63.68}} = 3.03 \text{ Hz}$$

$$\text{Unsprung Natural Frequency, } f_u = \frac{1}{2\pi} \sqrt{\frac{28.03 \times 10^3 + 128.7 \times 10^3}{20.23}} = 14.01 \text{ Hz} \quad (2)$$

$$\text{Sprung Critical Damping, } C_{crit} = 4\pi m_s f_s R_m = 5261.6 \text{ N/m/s}$$

$$\text{Selected Damping Coefficient, } C = 0.7 C_{crit} = 3683 \text{ N/m/s (0 d.p.)}$$

$$\text{Unsprung Critical Damping, } C_{crit} = 4\pi m_u f_u R_m = 7728.6 \text{ N/m/s}$$

$$\text{Unsprung Damping Coefficient} = \frac{C}{C_{crit}} = \frac{3683}{7728.6} = 0.477 \text{ (3 d.p.)}$$

$$\text{Low Frequency Bump Damping} = \frac{2}{3}C = 2455 \text{ N/m/s (0 d.p.)}$$

$$\text{Low Frequency Rebound Damping} = \frac{3}{2}C = 5525 \text{ N/m/s (0 d.p.)}$$

$$\text{High Frequency Bump Damping} = 0.5 \times 2757 = 1228 \text{ N/m/s (0 d.p.)}$$

$$\text{High Frequency Rebound Damping} = 0.5 \times 6204 = 2762 \text{ N/m/s (0 d.p.)}$$

3.3 Damping Curves

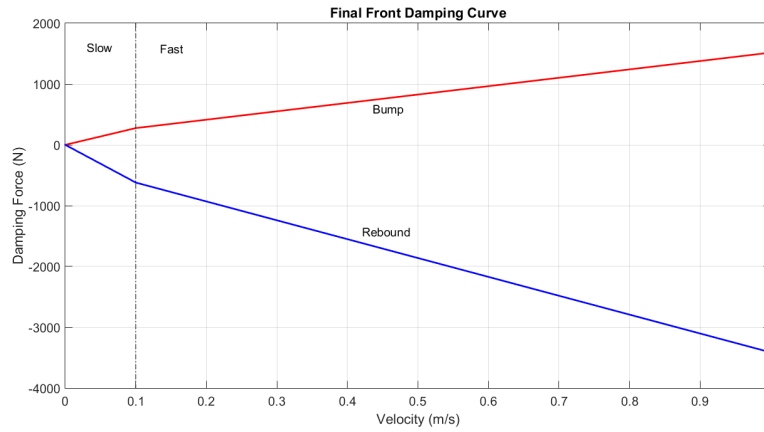


Figure 18: Front Suspension Damping Curve

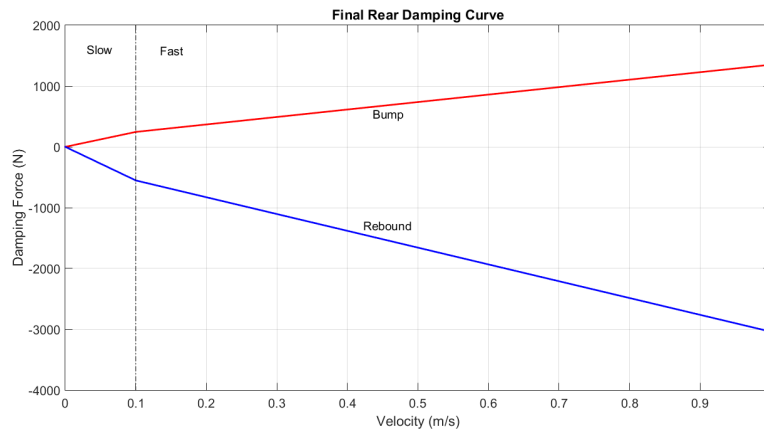


Figure 19: Rear Suspension Damping Curve



3.4 Appendix C: Suspension Load Calculations

3.4.1 Front Suspension Loads

$$\text{Max Vertical Load} = 1120N$$

$$\text{Max Longitudinal Load} = 1800N$$

$$\text{Max Lateral Load} = 2810N$$

Maximum Vertical Load Case

Assuming small angles for the wishbones:

$$W_{vert} = F_{Pullrod} \times \sin\theta$$

$$F_{pullrod} = \frac{W_{vert}}{\sin\theta} = \frac{1120}{\sin(51.6)} = 1429N$$

$$H_{pushrod} = F_{pullrod} \times \cos\theta = 1429\cos(51.6) = 888N$$

Taking horizontal components of force at lower wishbone mounting point:

$$888 = F_{AC}\cos(22.5) + F_{AD}\cos(22.5)$$

$$\therefore F_{AC} = 481N \quad F_{AD} = 481N$$

Maximum Cornering Case

During Cornering, the vertical load case is calculated as above, the horizontal component is evaluated by taking moments about A:

$$F_{top} \times h_2 = W_{lat} \times h_1$$

$$\therefore F_{top} = \frac{h_1}{h_2} \times W_{lat} = \frac{113.90}{275.20} \times 2810 = 1163N$$

$$F_{upper\ wishbone} = \frac{F_{top}}{\cos(16.31)} = \frac{1163}{\cos(16.31)} = 1116N$$

Taking horizontal components at upper wishbone mounting point:

$$1116 = F_{EG}\cos(22.5) + F_{EF}\cos(22.5)$$

$$\therefore F_{EG} = 604N \quad F_{EF} = 604N$$

Taking moments about node E gives:

$$F_{bottom} \times h_2 = W_{lat} \times (h_2 + h_1)$$

$$\therefore F_{bottom} = \frac{h_1 + h_2}{h_2} \times W_{Lat} = \frac{113.90 + 275.20}{275.20} \times 2810 = 3973N$$

$$F_{lower\ wishbone} = \frac{F_{bottom}}{\cos(13.52)} = \frac{3973}{\cos(13.52)} = 4086N$$

Taking horizontal components at lower wishbone mounting point:

$$4086 = -F_{AC}\cos(22.5) - F_{AD}\cos(22.5)$$

$$\therefore F_{AC} = -2228N(\text{compression}) \quad F_{AD} = -2228N(\text{compression})$$

Combining the Vertical Load and Cornering load case gives the worst case loading on the suspension arms giving final loads on the members as:

$$F_{AC} = -1744N(\text{compression}), F_{AD} = -1744N(\text{compression}), F_{EG} = 608N, F_{EF} = 608N, F_{pullrod} = 1429N$$



Using the work of Seward [4] it is possible to size steel bars to take these loads with a 50% safety factor to take into account kerb strikes and unforeseen contact during the dynamic events. It was assumed the components will be manufactured with 4130 steel ($\sigma_y = 350 \text{ N/mm}^2$, $E = 200000 \text{ N/mm}^2$):

$$\text{min. Area, } A_{\text{Upper wishbone}} = \frac{1.5 \times F_t}{\sigma_y} = \frac{912}{350} = 2.60 \text{ mm}^2$$

hence from [4] we can choose a 12mm diameter 1.5mm thickness, 4130 steel bar to ensure no failure of the upper wishbone occurs.

For the lower wishbone in compression we can determine a minimum size using the Euler buckling load of the member:

$$\text{Required } I = \frac{F_c \times 1.5L^2}{\pi^2 E} = \frac{1744 \times 1.5 \times 300^2}{\pi^2 \times 200000} = 119.3 \text{ mm}^4$$

Again a 1.5mm thick bar can be used with 15mm diameter comprised of 4130 steel. The data from [4] shows this will sustain the dynamic loads acting on the vehicle.

For 4130 Steel (density 7850 kg/m^3) this gives:

$$\text{Mass}_{\text{UpperWishbone}} = 7850 \times 10^{-9} \times 2 \times 12\pi \times 1.5 \times 300 = 0.266 \text{ kg}$$

$$\text{Mass}_{\text{LowerWishbone}} = 7850 \times 10^{-9} \times 2 \times 15\pi \times 1.5 \times 300 = 0.333 \text{ kg}$$

$$\text{Mass}_{\text{Pullrod}} = 7850 \times 10^{-9} \times 12\pi \times 1.5 \times 430.54 = 0.191 \text{ kg}$$

$$\text{Mass}_{\text{Rocker}} = 0.090 \text{ kg} \quad \text{Mass}_{\text{Spring/Damper Assembly}} = 0.280 \text{ kg}$$

$$\text{Mass}_{\text{Front Assembly}} = 0.266 + 0.333 + 0.191 + 0.090 + 0.280 = 1.16 \text{ kg}$$

3.4.2 Rear Suspension Loads

$$\text{Max Vertical Load} = 1298 \text{ N}$$

$$\text{Max Longitudinal Load} = 1800 \text{ N}$$

$$\text{Max Lateral Load} = 3180 \text{ N}$$

Maximum Vertical Load Case

$$F_{\text{pullrod}} = \frac{W_{\text{vert}}}{\sin\theta} = \frac{1298}{\sin(47.7)} = 1755 \text{ N}$$

$$H_{\text{pushrod}} = F_{\text{pullrod}} \times \cos\theta = 1755 \cos(47.7) = 1181 \text{ N}$$

Taking horizontal components of force at lower wishbone mounting point:

$$1181 = F_{AC} \cos(23.5) + F_{AD} \cos(23.5)$$

$$\therefore F_{AC} = 644 \text{ N} \quad F_{AD} = 644 \text{ N}$$

Maximum Cornering Case

During Cornering, the vertical load case is calculated as above, the horizontal component is evaluated by taking moments about A:

$$\therefore F_{\text{top}} = \frac{h_1}{h_2} \times W_{\text{lat}} = \frac{118.90}{270.20} \times 3180 = 1399 \text{ N}$$

$$F_{\text{upper wishbone}} = \frac{F_{\text{top}}}{\cos(7.93)} = \frac{1399}{\cos(7.93)} = 1413 \text{ N}$$

Taking horizontal components at upper wishbone mounting point:

$$1413 = F_{EG} \cos(23.5) + F_{EF} \cos(23.5)$$

$$\therefore F_{EG} = 770 \text{ N} \quad F_{EF} = 770 \text{ N}$$



Taking moments about node E gives:

$$\therefore F_{bottom} = \frac{h_1 + h_2}{h_2} \times W_{Lat} = \frac{118.90 + 270.20}{270.20} \times 3180 = 4579N$$

$$F_{lower wishbone} = \frac{F_{bottom}}{\cos(7.01)} = \frac{4579}{\cos(7.01)} = 4614N$$

Taking horizontal components at lower wishbone mounting point:

$$4614 = -F_{AC}\cos(23.5) - F_{AD}\cos(23.5)$$

$$\therefore F_{AC} = -2516N(\text{compression}) \quad F_{AD} = -2516N(\text{compression})$$

Combining the Vertical Load and Cornering load case gives the worst case loading on the suspension arms giving final loads on the members as:

$$F_{AC} = -1746N(\text{compression}), F_{AD} = -1746N(\text{compression}), F_{EG} = 770N, F_{EF} = 770N, F_{pullrod} = 1755N$$

The compression load on the lower wishbone is identical to before hence also need a 1.5mm thick bar with 15mm diameter comprised of 4130 steel. For the tensile load on the upper wishbone:

$$\text{min. Area, } A_{Upper wishbone} = \frac{1.5 \times F_t}{\sigma_y} = \frac{1155}{350} = 3.30mm^2$$

hence from [4] we can choose a 12mm diameter 2mm thickness, 4130 steel bar to ensure no failure of the upper wishbone occurs. The pullrod loads can also be satisfied by such a bar. Hence a final assembly mass can be calculated:

$$Mass_{UpperWishbone} = 7850 \times 10^{-9} \times 2 \times 12\pi \times 2 \times 380 = 0.449kg$$

$$Mass_{LowerWishbone} = 7850 \times 10^{-9} \times 2 \times 15\pi \times 1.5 \times 380 = 0.422kg$$

$$Mass_{Pullrod} = 7850 \times 10^{-9} \times 12\pi \times 2 \times 410.59 = 0.243kg$$

$$Mass_{Rocker} = 0.090kg \quad Mass_{Spring/Damper Assembly} = 0.280kg$$

$$Mass_{Front Assembly} = 0.449 + 0.422 + 0.243 + 0.090 + 0.280 = 1.48kg$$

3.6 Appendix E: CFD Wind Speed Validation

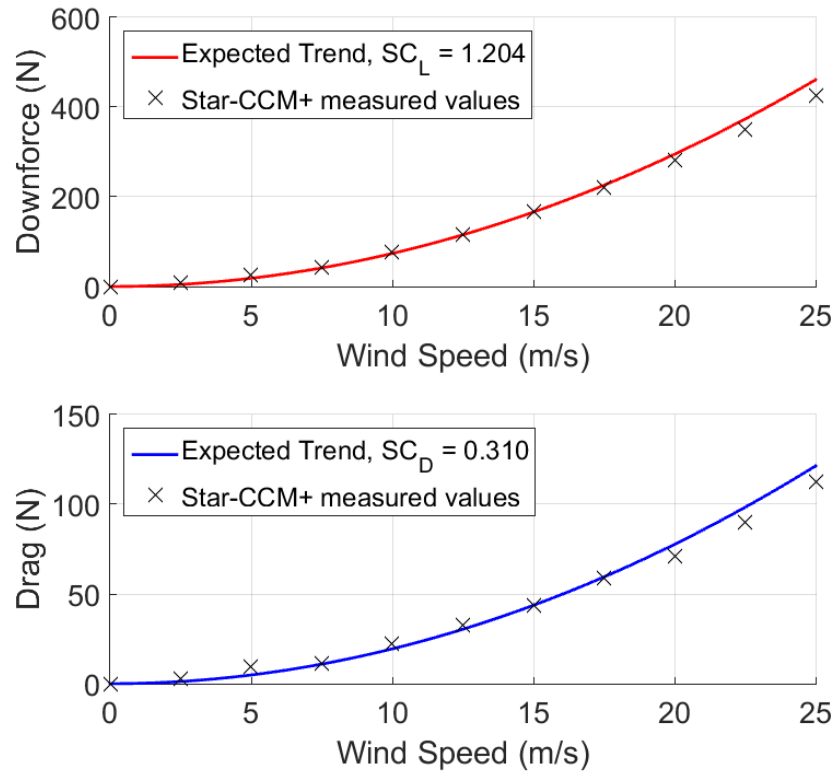


Figure 22: Comparison of expected trend of drag & downforce for the initial wing concept to values measured from Star-CCM+

3.7 Appendix F: Rear Wing Regulation Envelope

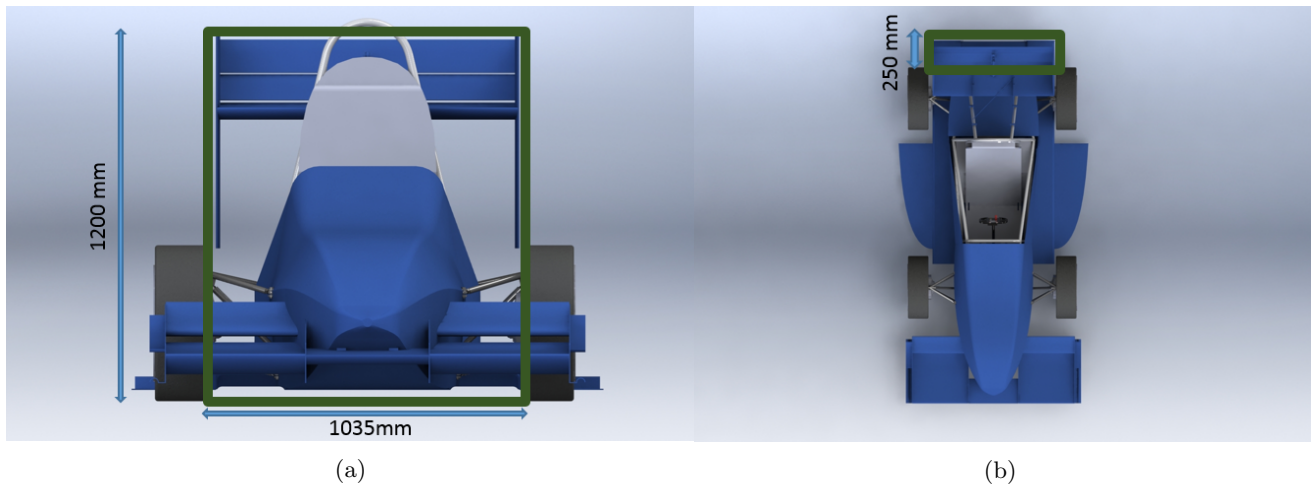


Figure 23: Region in which the rear wing can be mounted by FSAE regulations

3.8 Appendix G: xFoil Aerofoil Data

3.8.1 Aerofoil Comparison

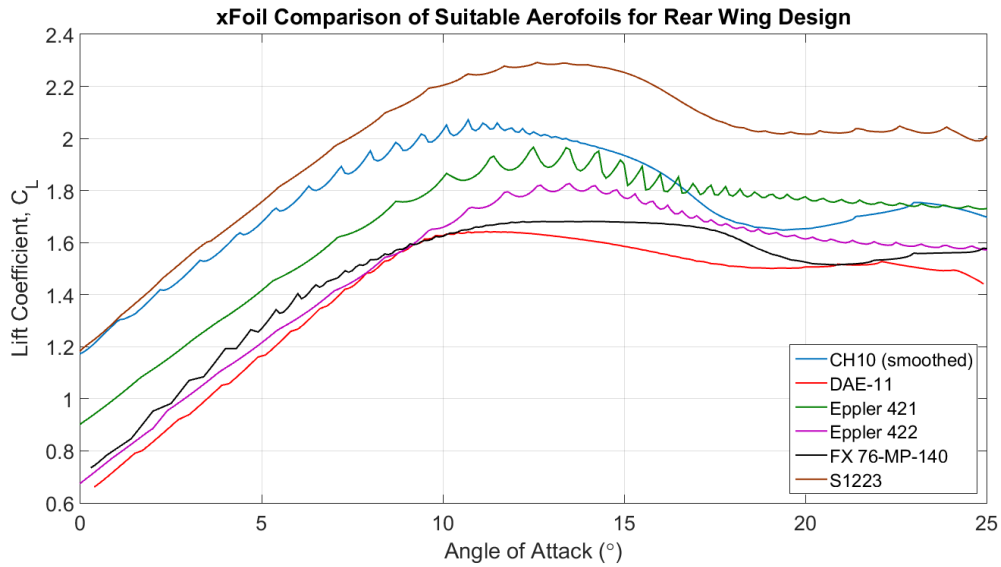


Figure 24: xFoil comparison of aerofoils designed for $Re = 250,000$

3.8.2 S1223

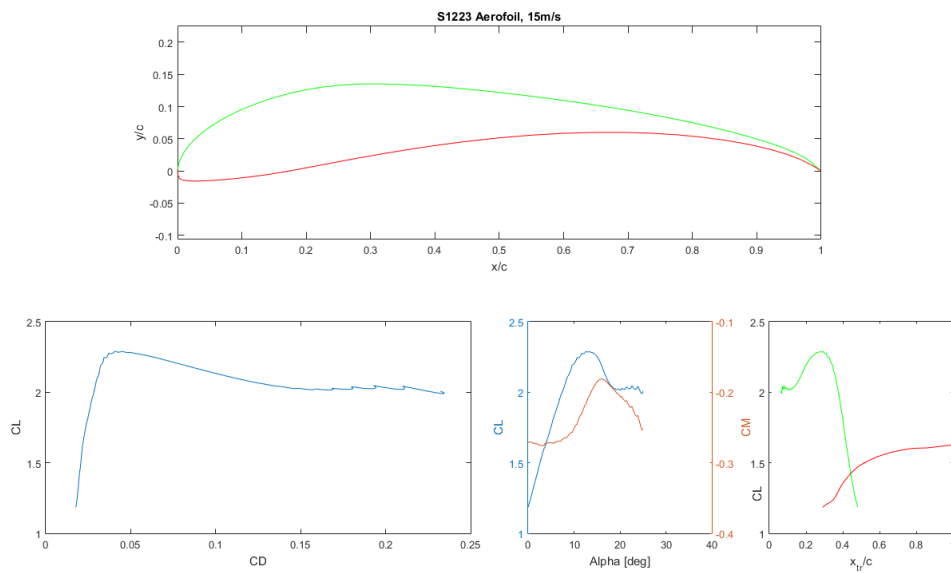


Figure 25: S1223 Aerofoil Drag Polar at 15 m/s produced via xFoil

3.8.3 DAE-11

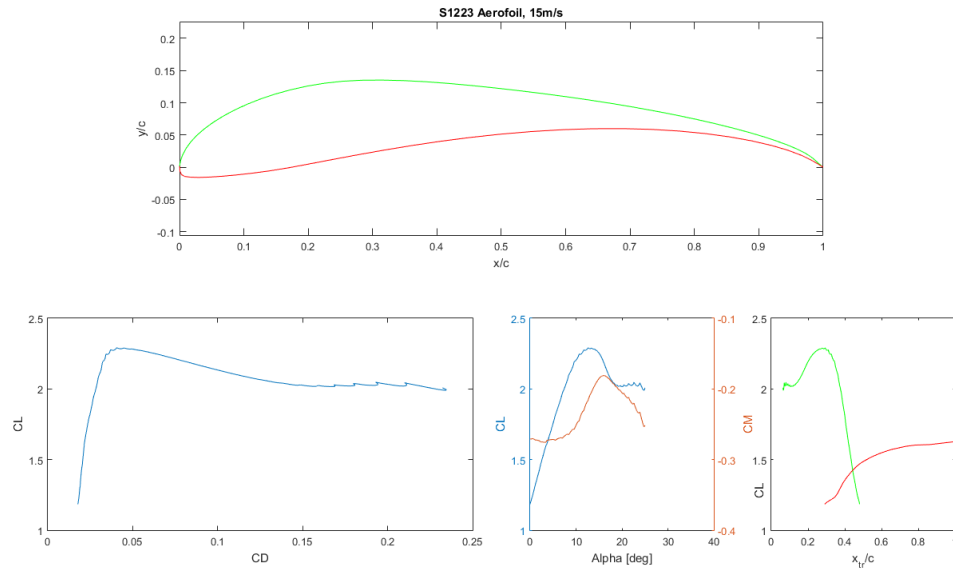


Figure 26: DAE-11 Aerofoil Drag Polar at 15 m/s produced via xFoil

3.9 Appendix H: Slot Gap Sweep

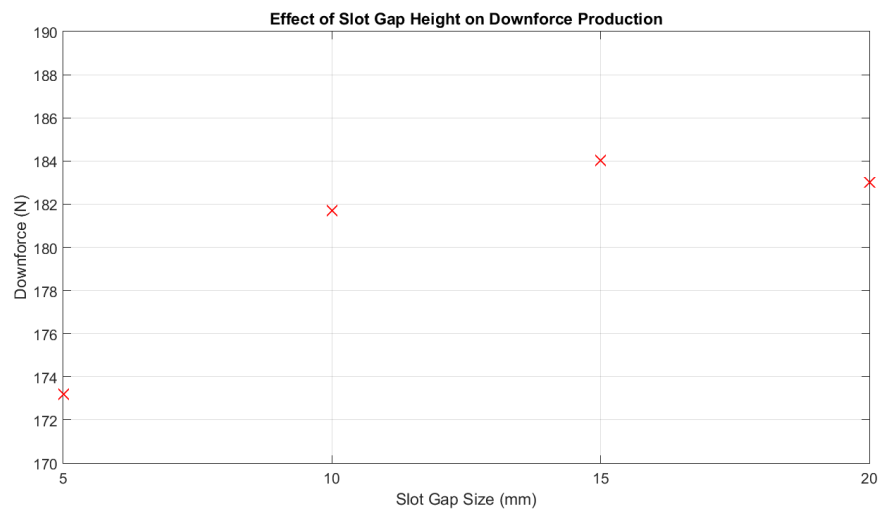


Figure 27: Graph summarising downforce produced by slot gap sweep CFD simulations